

Data Types in Python

- 1) Basic data types —
int, float, complex, bool, string
- 2) Container data types —
list, tuple, set, dictionary
- 3) User defined data types —
class

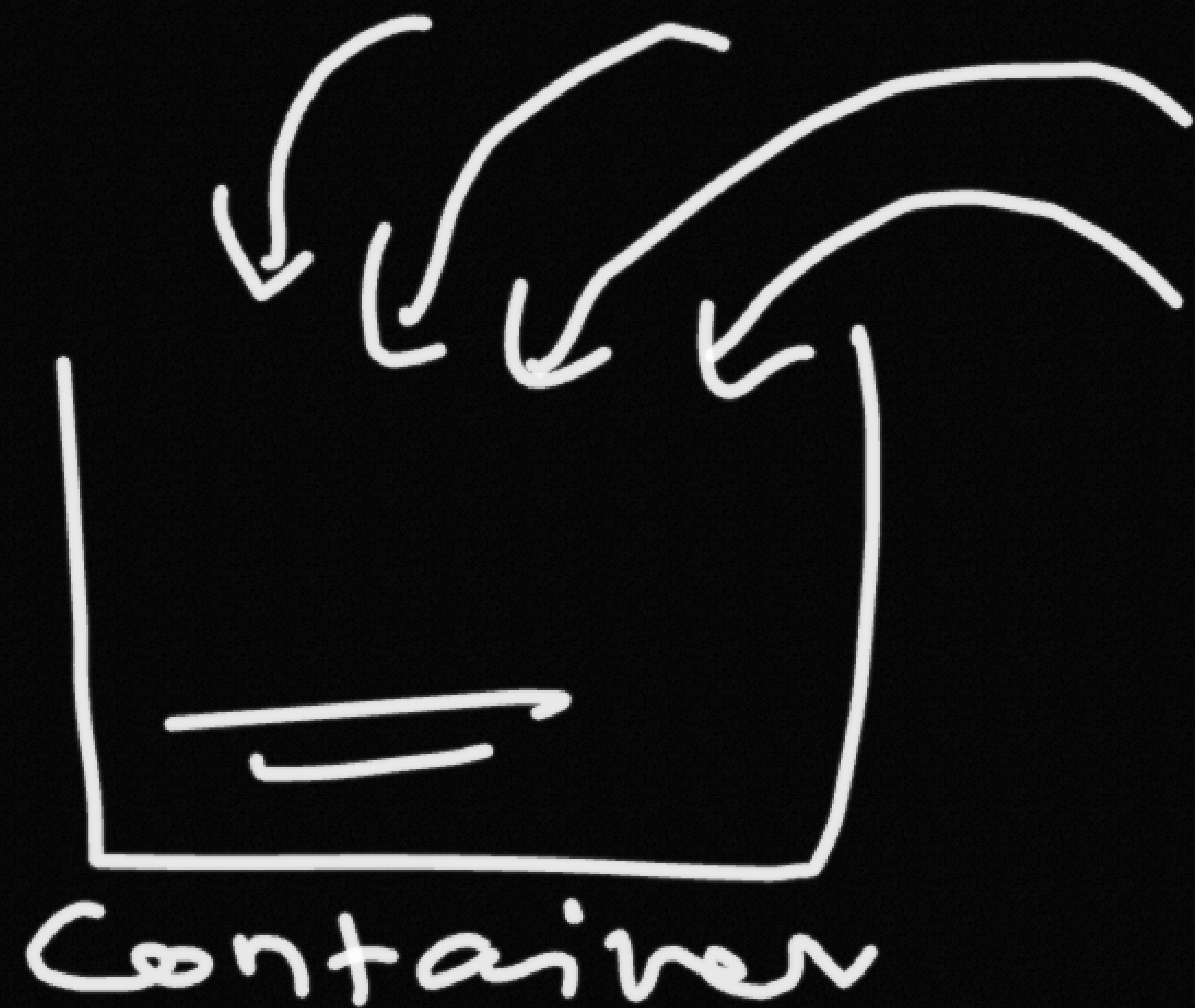
List -

1) List is One of the Container data type in python.

Basic data types

age = 25

name = "g_nesh"



List syntax:

List_name = [ele1, ele2, ele3, ...]

list1 = [10, 10.1, 10+2j, "gareh", True]

List is represented using square bracket & elements are separated by comma.

- Homogenous elements -
all the elements are of
same data type.

10, 20, 30, 40, 50
| | | | |

"ganeesh", "mahesh", "surath", "ajeth"

Heterogenous element -
all the elements are
different data types.

10, 10+2j, 10.1, "gareesh", True

Duplicate elements are
allowed in list

list 3 = [10, 10.1, "gareeb",
10, 10.1]

Repetition of elements using
operator.

list 4 = [10, 20, 30] * 3

o/p - [10, 20, 30, 10, 20, 30,
10, 20, 30]

Indexing - if we want to access individual element of the list we use indexing.

There are two types

1) Positive indexing - starts from begg.
4 0

2) Negative indexing - starts from end
- 1

Positive ind	0	1	2	3	4
list1	{10	20	30	40	50}
Negative ind	-5	-4	-3	-2	-1 ←

Print (list1) → [10, 20, 30, 40, 50]
 Print (list1[0]) → 10 ← Print (list1[-5])
 Print (list1[1]) → 20 ← Print (list1[-4])
 Print (list1[2]) → 30 ← Print (list1[-3])
 Print (list1[3]) → 40 ← Print (list1[-2])
 Print (list1[4]) → 50 ← Print (list1[-1])

slicing - Slicing is used to access sublist (group of elements)

`list_name[start; stop+1; step]`

default values → 0 len(list) 1

 (5)

Pos. Ind	0	1	2	3	4
list2	10	20	30	40	50

print(list2) → [10, 20, 30, 40, 50]

print(list2[0:3:1]) → [10, 20, 30]

print(list2[1:4:1]) → [20, 30, 40]

print(list2[2:5:1]) → [30, 40, 50]

print(list2[0:2:1]) → [10, 20]

the ind
list 2

0	1	2	3	4
10	20	30	40	50

print(list2[0:5:1]) → [~~10~~, 20, ~~30~~, ~~40~~, ~~50~~]

print(list2[0:1:1]) → 10

print(list2[0:5:2]) → [10, 30, 50]

print(list2[0:5:3]) → [10, 40]

the ind

	0	1	2	3	4
list 2	10	20	30	40	50

`print(list2[:4])` — `[10, 20, 30, 40]`

`print(list2[2:])` — `[30, 40, 50]`

`print(len(list2))` — `5`

`print(list2[:3])` — `[10, 20, 30, 40, 50]`

`print(list2[0:5:1])` — `[10, 20, 30, 40, 50]`

iterate over list using for
loop.

list3 = [10, 20, 30, 40, 50]

for var in list3:

 print(var)

==
o/p - 10
 20
 30
 40
 50

while loop - ②

list 4 = [10, 20, 30]

index = 0

while (index < len(list 4))

print [list 4 [index]]

index = index + 1

1
2 ← 0 + 1
3 ← 1 + 1
4 ← 2 + 1

o/p - 10
20
30

0 < 3

True/False

1 < 3 ✓

2 < 3 ✓

3 < 3 ✗

index

enumerate() - enumerate is one of the built in function in Python. we can print element along with its index.

Syntax :-

```
for index, var in enumerate(list1):  
    print(index, var)
```

```
list 1 = [10, 20, 30, 40, 50]
```

```
for var in list1:
```

```
    print(var)
```

o/p -

- 10
- 20
- 30
- 40
- 50

list 2 = [10, 20, 30, 40, 50]

for index, var in enumerate(list2):

 ↑ print(index, var)

o/p -

0	10
1	20
2	30
3	40
4	50

